National Institute of Open Schooling
Senior Secondary Course: Mathematics
Lesson 9: Quadratic Equations and Linear Inequalities
Worksheet-9

1. Solve the quadratic equation
$\sqrt{2}\,x^2 + x + \sqrt{2} = 0$

2. Examine the nature of roots in each of the following quadratic equations and verify them by formula.
   (i) $x^2 + 5x + 12 = 0$
   (ii) $2x^2 - 3x + 12 = 0$

3. Develop the quadratic equations whose roots are
   (i) $2 \text{ and } -2$
   (ii) $-3 \text{ and } 7$

4. If a and b are the roots of the equation $x^2 + 5x + 5 = 0$, the find the value of $(a^2 + b^2)$.

5. If the quadratic equation $x^2 - 6x + P = 0$ has equal roots, then find the value of $P + \frac{1}{P}$.

6. If p and q be the roots of the equation $2x^2 - 8x + 6 = 0$, form a quadratic equation whose roots are $p^2 \text{ and } q^2$.

7. Represent the linear inequality $4x - 16 \geq 0$ graphically.

8. Sole the linear inequality for real x
$3(x-1) \leq 3x - 5$

9. Solve and show on the number line of the linear inequality
$5x - 3 \geq 3x - 5$

10. Shyam obtained 35 and 40 marks in first two unit tests. Find maximum marks, he should get in the third test to have an average of at least 30 marks.